

GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027
CURRENT AFFAIRS DECODED
SUMMARY NOTES
19th July to 25th July, 2026

Article : Finning - A Cruel, Wasteful Practice

Finning involves removing a shark's fins and casting the rest of the shark overboard, often alive, causing it to subsequently drown.

- Illegal shark fins have been repeatedly landed in Senegal's capital Dakar, one of Africa's busiest fishing ports, by dozens of Chinese and Taiwanese longline tuna vessels, per a report by the Environmental Justice Foundation.
- Senegal is party to the **International Commission for the Conservation of Atlantic Tunas (ICCAT)**, which bans retention of certain shark species and caps fins at 5% of overall shark weight onboard, hard to enforce in practice.

Why Sharks Matter Ecologically

- As apex predators, sharks regulate marine food webs and maintain ocean ecosystem balance; their slow reproduction rates make populations highly vulnerable to overfishing.

What Is IUU Fishing?

Illegal, Unreported and Unregulated (IUU) fishing is activity that violates laws, goes unreported to authorities, or occurs outside regulatory frameworks, a major driver of global fisheries depletion.

Article : Shah Unveils India's First Immersive Language Museum

Union Home Minister Amit Shah inaugurated the **Museum of Word (Shabdlok)** at the National Library campus in Kolkata, calling it a step to preserve India's virasat (heritage).

- Developed by the National Library under the Union Culture Ministry, it is India's first immersive language museum, celebrating linguistic heritage through interactive, technology-driven exhibits.
- Its first phase chronicles the diversity of Indian languages, scripts, and literature; Shabdlok celebrates India's 22 official languages, tracing their journey from oral traditions to modern electronic texts.
- India's 22 Scheduled Languages are listed in the **Eighth Schedule** of the Constitution, officially recognised for purposes including translation of official documents and competitive exams.

Article : India's F&O Boom Needs Adequate Protections

Derivatives are financial contracts between two or more parties that derive their value from an underlying asset, group of assets, or benchmark, such as shares, bonds, currencies, or commodities. They are used to hedge risks arising from future price fluctuations, can be exchange-traded or Over the Counter (OTC), and are regulated by SEBI.

- **Forwards:** customised contracts between two parties to buy or sell an asset at a specified price on a future date; traded outside the stock exchange (OTC).
- **Futures:** standard agreements to buy or sell a commodity, asset, or security at a predetermined price on a specified future date; standardised for quality and quantity and traded on futures exchanges.
- **Options:** give the holder the right, but not the obligation, to buy or sell an underlying security at a predetermined "strike price."

The Problem

India's Futures and Options (F&O) trading boom has led to devastating financial losses for retail investors, with mounting debt pressure recently linked to multiple suicide cases.

- A July 2025 SEBI study found over 90% of individual traders incurred heavy losses over FY22-FY25; aggregate retail losses surpassed ₹2.8 lakh crore, with only 1% of retail traders making over ₹1 lakh profit.
- The "rigged playing field": 97% of institutional profits and 96% of proprietary trading profits come from algorithmic trading using exchange-co-located code, systems retail traders cannot access.
- Institutions primarily use derivatives to hedge existing portfolios against market risk, unlike many retail participants who use them for speculative, leveraged bets on short-term price movements.
- **Algorithmic and high-frequency trading**: automated trading using computer algorithms to execute orders at extremely high speeds, often co-located within exchange infrastructure for latency advantages.
- Global regulatory models: Singapore's Monetary Authority requires formal Customer Knowledge Assessments; the US's FINRA enforces Know-Your-Customer and Suitability Rules, making brokers legally liable gatekeepers.

Article : Making Sense of Embodied AI

- **Embodied AI** is AI that has a physical body and can perceive, move, and interact with its environment to perform tasks autonomously.
- **Neuromorphic AI** is AI designed to work like the human brain by mimicking the way neurons and synapses process information.

Article : India, Spain Advance Talks on UPI-Bizum Interoperability

Digital Public Infrastructure (DPI) refers to a collection of shared digital platforms and systems operating at the scale of an entire society, designed to be secure, interoperable, and built on open standards, enabling public and private entities to provide services efficiently.

- **UPI** is an advanced version of the Immediate Payment Service (IMPS): a round-the-clock, real-time payment system facilitating inter-bank peer-to-peer (P2P) and person-to-merchant (P2M) transactions.
- Developed by the **National Payments Corporation of India (NPCI)**, an umbrella organisation for retail payments and settlement systems, initiated by the RBI and the Indian Banks' Association (IBA); it is a "Not for Profit" company under Section 8 of the Companies Act, 2013.
- UPI has transformed India "from queues to QR codes," emerging as the backbone of the country's digital financial ecosystem, and uses two-factor authentication.
- **BHIM (Bharat Interface for Money)**: a mobile payment app developed by NPCI, letting users send and receive money instantly using UPI.
- **Global acceptance**: Bhutan, France, Mauritius, Nepal, Singapore, Sri Lanka, UAE, Qatar, and Cambodia.

MDR (Merchant Discount Rate)

MDR is a fee charged to merchants by banks for accepting customer payments via credit/debit cards or QR code.

- In December 2019, the Government decided to waive MDR charges on transactions done through RuPay and BHIM-UPI to push digital payments, effective from 1 January 2020.

Article : Four Indian Seafarers Killed Near Odessa

Four Indian citizens were killed and one was critically injured in an attack on the merchant vessel **MV Golden Leo** near Ukraine's port of Odessa on Sunday, 19 July 2026.

- The vessel was attacked while departing the port of Odessa; 17 crew members were on board, including five Indian nationals. India's Ministry of External Affairs confirmed four Indian nationals had died, while one was hospitalised in critical condition.
- India condemned the incident and called such attacks impeding freedom of navigation "deplorable," reiterating its opposition to attacks on merchant vessels as violations of freedom of navigation.
- Ukraine's Acting Foreign Minister, Andrii Sybiha, blamed Russia for the attack and said five crew members remained missing. This is the first instance of Indian seafarers being killed in the Russia-Ukraine war, which began in February 2022.
- The incident occurred near Odessa, Ukraine's largest sea port on the Black Sea.

Mediterranean Sea

Bordered by 21 countries: Morocco, Algeria, Tunisia, Libya, Egypt, Israel, Lebanon, Syria, Turkey, Greece, Albania, Montenegro, Bosnia and Herzegovina, Croatia, Slovenia, Italy, France, and Spain, with Cyprus and Malta as its main islands.

- Connected to the Atlantic Ocean in the west through the Strait of Gibraltar, to the Black Sea in the east through the Dardanelles Strait, and to the Red Sea in the south through the Suez Canal. Africa's Nile River drains into it.

Adriatic Sea

An arm of the Mediterranean Sea lying between Italy and the Balkan Peninsula, extending from the Gulf of Venice south to the Strait of Otranto, linking it to the Ionian Sea. Bordering countries: Italy, Croatia, Bosnia and Herzegovina, Montenegro, Albania, and Slovenia.

Black Sea

Also known as the **Euxine Sea**, one of the world's major inland seas. Its bordering countries can be remembered as **BURGeR-T**: Bulgaria, Ukraine, Russia, Georgia, Romania, and Turkiye.

- Linked to the Sea of Marmara through the Bosphorus Strait, and then to the Aegean Sea (an elongated embayment of the Mediterranean) through the Dardanelles Strait; the Turkish straits system (Dardanelles, Bosphorus, and the Marmara Sea) forms a transitional zone between the Mediterranean and the Black Sea.
- Also connected to the Sea of Azov by the Strait of Kerch. Surrounded by the Pontic Mountains to the south, the Caucasus to the east, and the Crimean Mountains to the north.

Caspian Sea

Earth's largest inland body of water by surface area, bordered by Russia (northwest), Azerbaijan (west), Iran (south), Turkmenistan (southeast), and Kazakhstan (northeast).

- In 2003, the Caspian littoral states signed the Tehran Framework Convention for the Protection of the Marine Environment of the Caspian Sea. It is an endorheic basin (a drainage basin with no outflow).
- **Baku**, the capital of Azerbaijan, is the largest city on the Caspian Sea.

Article : James Webb Telescope - Beta Pictoris b: Hide and Seek

The **James Webb Space Telescope (JWST)** is a monumental leap in space exploration and astrophysics. Launched on 25 December 2021, it is the largest and most powerful space telescope ever built, serving as the premier space-based observatory for the next decade.

- Developed by NASA with contributions from the European and Canadian space agencies, JWST observes the universe primarily in the infrared spectrum, conceived as the next-generation successor to the Hubble Space Telescope.

Key Design Features

- **Infrared optimisation:** designed primarily for infrared astronomy, observing objects too old, distant, or faint for visible-light detection.
- **Large primary mirror:** a 6.5-metre diameter primary mirror capturing more light for higher-resolution images, made of 18 hexagonal, gold-coated beryllium segments that fold for launch and unfold in space.
- **Sunshield protection:** a five-layer sunshield blocking solar light and heat, keeping instruments at the extremely low temperatures needed for infrared observation.
- **Location at Lagrange Point 2 (L2):** operates from a stable orbit around the Sun-Earth L2 point, minimising light interference and reducing fuel needed for orbital corrections.
- Highly sensitive instruments enable the study of the earliest stars and galaxies formed after the Big Bang, aided by a passive cryogenic cooling system reaching temperatures as low as 40 Kelvin (-233°C), and precision guidance sensors for accurate, stable long-duration observation.

Lagrange Points

Lagrange points are special positions in space where objects tend to stay in a stable or nearly stable position; spacecraft placed there need very little fuel to remain in place, making them useful for satellites, telescopes, and observatories.

- Of the five Lagrange points, three (L1, L2, L3) are unstable and lie along the line connecting the two large masses; two (L4, L5) are stable, forming the apex of equilateral triangles with the large masses at their vertices.

JWST vs Hubble

Feature	JWST	Hubble
Wavelength focus	Optimised for infrared astronomy, seeing through dust clouds and viewing redshifted light of distant galaxies	Predominantly visible and ultraviolet light
Primary mirror	6.5 metres in diameter	2.4 metres in diameter
Light-gathering capability	Far more, enabling detection of fainter, more distant objects	Less, compared to JWST
Location	Second Lagrange point (L2): stable vantage point, uninterrupted sky observation	Low Earth orbit: less stable than L2

Benefits of JWST

- Advancing astrophysical knowledge, exoplanet exploration, educational and scientific collaboration, and acting as a catalyst for future space missions.

Article : Prevention of Insults to National Honour Act, 1971

Enacted to penalise acts of disrespect to the National Flag, the Constitution, and the National Anthem.

- **Section 3** specifically criminalises preventing the singing of the national anthem

or causing disturbance to an assembly engaged in singing it.

Constitutional Basis

- **Article 51A(a)** places a Fundamental Duty on every citizen to respect the National Flag and National Anthem; the National Song finds no mention.
- **Article 19(1)(a)** (freedom of speech and expression) and **Article 25(1)** (freedom of conscience and religion) are both invoked in cases balancing compulsion to sing against individual rights.

History of Vande Mataram

Written by **Bankim Chandra Chattopadhyay**, first published in his 1882 novel *Anandamath*. India's Constitution does not explicitly mention a national song.

Article : Building an Atmanirbhar Philanthropy Ecosystem

Domestic private giving through family philanthropy, Corporate Social Responsibility (CSR), and individual donors has grown rapidly and now far outstrips annual foreign philanthropic inflows.

- Domestic private philanthropy now exceeds ₹1.18 lakh crore a year, more than five times foreign inflows, according to the Bain-Dasra India Philanthropy Report 2026.

Corporate Social Responsibility (CSR)

- India's CSR framework is established by **Section 135** of the Companies Act, 2013, mandating eligible companies to contribute towards societal development.
- **Eligibility:** applies to companies with a net worth of ₹500 crore or more, an annual turnover of ₹1,000 crore or more, or a net profit of ₹5 crore or more in the previous financial year.
- **Spend requirement:** eligible companies must allocate at least 2% of their average net profit from the previous three years towards CSR activities.

Permitted CSR Activities (Schedule VII)

- Eradication of hunger, poverty, and malnutrition; healthcare, sanitation, and safe drinking water.
- Promotion of education, vocational skills, and livelihood enhancement.
- Gender equality, women's empowerment, and support for senior citizens and vulnerable groups.
- Environmental sustainability, ecological balance, and conservation of natural resources.
- Protection and promotion of national heritage, art, culture, and traditional crafts.
- Welfare of armed forces, CAPF and paramilitary veterans and their dependents.
- Promotion of rural, national, Paralympic, and Olympic sports.
- Contributions to the Prime Minister's relief and other government welfare funds.
- Support to research, innovation, incubators, and public-funded institutions aligned with the SDGs.

Article : Buried Questions - The Teesta Stage-VI Tunnel Tragedy

The Teesta River

- Originates from Tso Lhamo Lake at an elevation of about 5,280 metres in North Sikkim, glacier-fed by sources like the Pahuni Glacier, Kangse Glacier, and ChhoLhamo Lake.
- A major tributary of the Brahmaputra River (known as the Jamuna in Bangladesh), flowing through India and Bangladesh.
- Major tributaries: Lachung Chhu, Dik Chhu, and Rangpo Chhu on the left bank, and the Rangit River on the right bank.
- About 83% of the basin lies in India, while 17% lies in Bangladesh.

Article : India Must Lead in Securing the Strait of Bab el-Mandeb

On 16 July 2026, Reuters reported that Iran had instructed the Houthis to close the Strait of Bab el-Mandeb, between Yemen and the Horn of Africa, if the United States attacked Iran's power plants, a risk to nearly one-third of India's entire trade.

Revisiting the 2024 Crisis

- When the Gaza war started in October 2023, the Houthis targeted at least 40 vessels in the Red Sea between November 2023 and March 2024, claiming the vessels were linked to or bound for Israel.
- The US and allies launched **Operation Prosperity Guardian** in December 2023 to ensure freedom of navigation, followed by **UN Security Council Resolution 2722** in January 2024; the Indian Navy rescued the crew of a British tanker set on fire that January.
- By the time the US declared a ceasefire in May 2025, the Houthis had been weakened, but much damage had already been done to global shipping and regional economies.

The Red Sea

An inlet of the Indian Ocean between Africa and Asia, a major hub for international tourism known for its diving spots and marine life. Bordering countries: Egypt, Saudi Arabia, Yemen, Sudan, Eritrea, and Djibouti.

- Connected to the Indian Ocean in the south through the Bab el-Mandeb Strait and the Gulf of Aden; its northern end features the Sinai Peninsula, the Gulf of Aqaba, and the Gulf of Suez, which leads to the Suez Canal.

Yemen

- Bordering countries: Saudi Arabia (north), Oman (east); bordering water bodies: the Red Sea (west), Gulf of Aden (south), Arabian Sea (southeast).
- Strategically located on Bab el-Mandeb, the strait linking the Red Sea and the Gulf of Aden. Yemen's Socotra Archipelago in the Indian Ocean is a UNESCO World Heritage Site.
- **Houthis**: a Zaidi Shia militant group from northwestern Yemen that emerged in the 1990s as a rebellion against the Yemeni government. Backed by Iran, they are part of the "Axis of Resistance" and target Israeli-linked ships in the Bab el-Mandeb Strait in response to the Gaza conflict.

Axis of Resistance

A coalition of Iranian-backed groups; the name is said to be inspired by former US President George W. Bush's use of the term "axis of evil," referring to Iran, Iraq, and North Korea.

- Members include **Hezbollah** (Shiite militant organisation in Lebanon), **Hamas** (Palestinian Sunni militant group in Gaza), and the **Houthis** (Zaydi Shia militant group in Yemen).

Article : Maintain Status Quo on Water Release, Says Cauvery Body

The **Cauvery (Kaveri) River** is designated as the "Dakshin Bharat ki Ganga," the Ganga of the South. Its basin extends over Tamil Nadu, Karnataka, Kerala, and the Union Territory of Puducherry, draining an area of 81 thousand sq km.

- The river rises at an elevation of 1,341 m at Talakaveri on the Brahmagiri range near Cherangala village of Kodagu (Coorg) district, Karnataka.

Key Tributaries and Structures

- **Hemavathi Dam**: a large masonry dam on the Hemavathi River (a Kaveri tributary) in Karnataka, critical for irrigation, drinking water, and hydroelectric power, serving Hassan and surrounding areas.
- **Laxman Tirtha River**: a Kaveri tributary originating in the Brahmagiri hills, Kodagu, Karnataka, flowing eastward and joining the Kaveri at the Krishna Raja

Sagara Lake.

- **Krishna Raja Sagar (KRS) Dam**: its reservoir is the main source of drinking water for all of Mysore city and almost the whole of Bengaluru.
- **Kabini River**: a major Kaveri tributary originating in Wayanad district, Kerala, at the confluence of the Panamaram and Mananthavady rivers.
- **Suvarnavathi**: a right-bank tributary; **Shimsha** and **Arkavathi**: left-bank tributaries. The proposed **Mekedatu project** sits at the confluence of the Kaveri and Arkavathi rivers.
- **Hogenakkal Waterfalls**, "the Niagara of India," lies on the Karnataka-Tamil Nadu border; the **Mettur Dam** is located at Mettur, Salem district, Tamil Nadu.
- **Bhavani River**: the second-largest river in Tamil Nadu, starting in the Nilgiri hills (Western Ghats), flowing through Kerala's Silent Valley National Park before turning back into Tamil Nadu; its tributaries **Noyal** and **Amaravati** join on the right bank. The Kaveri then passes through Tiruchirappalli.
- The **Grand Anicut (Kallanai)**: one of the oldest and most remarkable irrigation structures in the world, a stone masonry dam on the Kaveri in Thanjavur district, Tamil Nadu, traditionally attributed to the Chola king Karikala Chola in the 2nd century CE.

Basin Share by State

- Tamil Nadu: 56%; Karnataka: 41%; Kerala: 3%.

Article : SC-Appointed Aravalli Panel's Consultation Process Excludes Rural People

The **Aravalli Range** is among the world's oldest living mountain systems and the geological spine of northwest India.

- Mountain type: an ancient fold mountain range, now eroded into residual hills with average elevations of 300-900 metres.
- Geographic extent: stretches about 692 km from Champaner in Gujarat, through Rajasthan and Haryana, to Raisina Hill in Delhi.
- Highest peak: **Guru Shikhar**, at 1,722 metres on Mount Abu, Rajasthan.

Article : Himachal Farmers Pioneer Sustainable Cultivation of the Endangered Herb Kutki

Farmers in Mandi, Himachal Pradesh, have adopted a sustainable cultivation model for the endangered medicinal herb **Kutki (Picrorhiza kurroa)**.

- The medicinal value lies in the dried roots and stems; because the useful part is underground, collectors traditionally uproot the whole plant, which is the root of the conservation problem.

Distribution

- Native to the alpine and sub-alpine Himalaya, roughly 2,700 to 4,500 m altitude, surviving in a very narrow altitude belt.
- In India: Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh; also found in Nepal, Bhutan, Pakistan, and Tibet (China), so its trade is international, which is why CITES applies.

The Sustainable Cultivation Model

Developed by the **Himalayan Research Group (HRG)** under the Department of Science and Technology (DST), farmers harvest only the **stolons** (horizontal stems), leaving the mother plant intact, enabling repeated harvesting, conserving the species, and providing a steady income.

- A stolon is a horizontal, creeping stem that grows new baby plants at its tip; since only the stolon is cut, the mother plant stays alive in the soil and can be harvested again and again, much like plucking tea leaves rather than cutting down the tea bush.

Conservation Status

- **IUCN Red List:** Endangered. **CITES:** listed under **Appendix II**, meaning international trade is allowed, but only with permits (Appendix I would mean trade is banned outright, Appendix II means trade is regulated, a common exam mix-up to avoid).

Medicinal Use

- Best known for its hepatoprotective (liver-protecting) properties, used in jaundice and liver disorders.

Article : What Came Out of the NATO Ankara Summit

Note: The 5% of GDP by 2035 defence spending target was agreed at NATO's 2025 summit in The Hague, not at Ankara in 2026. Ankara's role was to have allies present concrete, credible national plans for reaching that already-agreed target, alongside reaffirming existing commitments.

- All 32 allies reaffirmed an "ironclad commitment" to collective defence under **Article 5**.
- Allies presented plans to reach the previously agreed target of 5% of GDP on defence and security-related spending by 2035 (3.5% on core defence requirements and 1.5% on broader defence-related spending).
- Continued support for Ukraine was reaffirmed.
- Members committed to building a high-technology, Europe-wide Defence Industrial Base.

How NATO's Spending Surge Affects Global Arms Buyers Like India

- As NATO spends more, the international arms market is shifting from a buyers' market to a sellers' market for the world's big arms buyers, including India.
- Example: US firm General Electric Aerospace has delayed supplying F-404 fighter jet engines, critically needed for the Indian Air Force's Tejas Light Combat Aircraft (LCA) programme.

About NATO

- Founded in 1949 by the Washington Treaty, also called the North Atlantic Treaty; headquartered in Brussels, Belgium, with its military headquarters, SHAPE, in Mons, Belgium.
- **32** members; newest members are Finland (2023) and Sweden (2024).
- Core principle: **Article 5**, collective defence, an attack on one is an attack on all. Article 5 has been invoked only once, after the 9/11 attacks in 2001.

GALLANT IAS